**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 115 · Two-Dimensional Arrays and Records

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. For s = 'COMPUTER', what is s.substring(1,3) in OCR ERL?

Q2. Which is an output from a greeting program?
A. A typed name
B. A mouse click
C. A displayed greeting
D. A key press

Q3. “Duplicating the same long code block everywhere always improves maintainability.” Correct this. Explain why.

Q4. Nested IF statements are hard to read. “Indentation proves the logic is correct.” Correct this. How does indentation help?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

Two-dimensional arrays — A 2D array stores values in rows and columns and can be accessed with two indexes.

Zero-based indexing — In a zero-indexed grid, the first row and column use index 0.

Records — A record groups related fields about one entity under a single item.

Choice of structure — Use a grid for table-like values and a record for named fields such as name and age.

## New learning

Database table: Each horizontal row is one pupil record. ID, Name and Mark are fields (columns).

Two-dimensional array: A fixed-size array emulates rows and columns; pupils[row,column] identifies one cell using zero-based indexes.

Data types: An OCR ERL array stores one element type. In a string-based table model, '72' is text; convert it before a numerical comparison.

Records: A named record can group an integer ID, text Name and integer Mark for one pupil without relying only on column positions.

## Worked example

Use the Pupils table: ID 101 Ada 72; ID 102 Ben 43; ID 103 Mia 81. In the string-based 2D array, what is pupils[0,1]? Explain the two indexes.

1. Label rows 0, 1 and 2 for Ada, Ben and Mia. Columns 0, 1 and 2 hold ID, Name and Mark.

2. The FIRST index 0 selects Ada's record (row 0), not column 0.

3. The SECOND index 1 selects the Name field (column 1).

4. pupils[0,1] is 'Ada'. Both indexes start at 0; marks are represented as text in this particular array.

# We Do · Practise with guidance

### Guided 1

Using the SAME Pupils table, identify the value in pupils[2,1] and explain the indexes. [3 marks]

Worked method: pupils[2,1] = 'Mia' [1]; index 2 selects third row [1]; index 1 selects Name column [1].

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

## Code First · Predict → Write → Test → Improve

The rows represent pupil records, with name in column 0 and attendance status in column 1. Complete the update to mark Ben present, then predict the output.

```
array pupils[2,2]
pupils[0,0] = "Ada"
pupils[0,1] = "Present"
pupils[1,0] = "Ben"
pupils[1,1] = "Absent"
pupils[1,1] = ...
print(pupils[1,1])
```

write

## Question 1

In the Pupils table, state the Name stored at row index 1, column index 1. [2 marks]

## Question 2

Explain what a row (record) and a column (field) represent in the Pupils table. [4 marks]

## Question 3

Describe one difference between a 2D array of text and a named pupil record. [3 marks]

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

## Exit Ticket · Answer before checking

A 2D array uses two indexes to locate an element.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

## Page 1 onwards · Do Now

Q1. OMP; indexing starts at zero and the second argument gives length.

Q2. C — A displayed greeting

Q3. Related logic can be reused, understood, tested and changed in one place.

Q4. Indentation shows blocks and nesting. Some languages require it. It helps people read and change code, but does not prove the logic is correct.

## I Do / We Do pages · Guided practice

Q1. pupils[2,1] = 'Mia' [1]; index 2 selects third row [1]; index 1 selects Name column [1].

## You Do pages · Code First

```
array pupils[2,2]
pupils[0,0] = "Ada"
pupils[0,1] = "Present"
pupils[1,0] = "Ben"
pupils[1,1] = "Absent"
pupils[1,1] = "Present"
print(pupils[1,1])
```

Expected result: Present

## You Do pages · Independent questions

Q1. Ben [1]; the first index selects row 1 and the second index selects Name/column 1 [1].

Q2. A row stores the data for one pupil [1] across that pupil's related values [1]. A field/column stores the same property, such as Mark [1], for all pupil records [1].

Q3. The 2D array is accessed by row and column indexes [1] and uses one consistent element type [1]; a named record uses field names and may store different field types [1].

## Exit Ticket · Model answer

True

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**